

OneVideo2Policy: A Resource-Constrained Reproduction of Synthetic Robot Learning from One Human Video

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Project Report

Abstract

Video2Robo proposes converting one monocular human demonstration into diverse robot training data using object tracking, 3D Gaussian Splatting (3DGS), and kinematically coupled robot motion. We study which parts of this claim survive on a single 6 GB consumer GPU without physical robots. Our pipeline uses SAM 2.1 Small and CoTracker3 for perception, evaluates TripoSR and Stable Fast 3D as lightweight reconstruction candidates, retains measured RGB-D geometry when both learned meshes fail a frozen proportion gate, and trains a 2.61M-parameter dual-view waypoint estimator for a measured ball-to-bowl task in robosuite. Across five training seeds and 250 paired combined-shift rollouts, the selected pipeline obtains $76.0 \pm 5.8\%$ mean success. A cumulative coverage ablation rises from 2% with 500 clean frames to 82% with 4,000 frames spanning Gaussian appearance, robot pose, camera, and lighting variation, although the five-seed mean remains below the frozen 80% combined-shift gate. Correspondence-constrained motion estimation reduces held-out image-plane error from 34.89 to 0.70 pixels, while single-image meshes retain 33–59% maximum secondary-dimension error. These results support synthetic visual diversity for simulated robustness, but also show that rendering fidelity, metric geometry, and closed-loop policy quality must be evaluated separately. We make no physical transfer claim.

1. Introduction

Robot imitation learning is limited by the cost of collecting diverse demonstrations. Video2Robo [3] addresses this bottleneck by recovering task-relevant motion from one RGB human video, editing a 3DGS scene, and rendering robot demonstrations under object, appearance, lighting, and viewpoint changes. Its evaluation uses six tasks, an NVIDIA L40 GPU, Diffusion Policy [2], and physical Franka arms. This leaves a practical question: what evidence can be reproduced on a small local machine when real robots and high-memory accelerators are unavailable?

We answer this question through a scoped reproduction

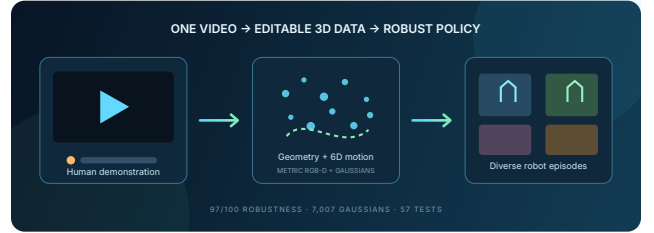


Figure 1. Hardware-free reproduction. One human sequence supplies masks, tracks, metric anchors, and object-relative motion. Measured geometry drives physics; 3DGS supplies appearance diversity; the learned component estimates a visual waypoint used by a validated phase controller.

rather than a claim of method equivalence. The running task is rigid-object placement using one HOI4D ball-to-bowl sequence [7]. The evaluation separates four links in the pipeline: perception, metric reconstruction, appearance generation, and closed-loop control. This separation matters because a visually plausible mesh may be unsafe for collision geometry, a higher-PSNR render may shift the policy input, and low offline action error may still yield zero closed-loop success.

Our study asks three questions. **RQ1:** can small perception and reconstruction models recover useful motion and geometry? **RQ2:** which forms of synthetic coverage improve robustness under combined visual and camera shift? **RQ3:** are the selected results stable across training seeds and increasing camera displacement? Our contributions are:

- a gated, reproducible video-to-simulation pipeline that runs on 6 GB VRAM;
- controlled reconstruction, tracking, Gaussian, policy, and resource measurements;
- five-seed paired evaluation and coverage/severity ablations; and
- explicit negative results for learned metric meshes and direct low-level visual policies.

2. Method and Evaluation Design

Perception and motion. We run SAM 2.1 Hiera Small [9] and offline CoTracker3 [5] over 72 frames. The source and target retain fixed identities; tracked pixels constrain frame-to-frame pose. For an asymmetric Record3D action-camera sequence, we compare mask centroid/area/principal axis fitting against correspondence-constrained 2D similarity fitting. Every third spatially sorted verified match is held out, giving 82 test correspondences over 13 frames.

Geometry and appearance. TripoSR [10] and SF3D [1] reconstruct an asymmetric object from one segmented RGB frame. After aligning each longest axis to an RGB-D reference, a mesh passes only if both secondary extents are within 15%. Failed learned meshes remain visual proposals; metric RGB-D and fitted primitives drive robosuite [11] physics. A sparse RGB-D 3DGS scene [6] is tuned on one frame and tested on three held-out views. Semantic anchors register simulator foreground to the Gaussian scene and depth ordering handles occlusion.

Policy. The task uses a 38.32 mm ball and a 99.12×56.93 mm bowl. The selected policy is a 2.61M-parameter dual-view CNN that predicts the ball waypoint from two 84×84 RGB views. A phase controller performs approach, grasp, transfer, release, and retreat. This hybrid design differs from Video2Robo’s end-to-end Diffusion Policy and is evaluated as learned visual localization plus scripted control. Direct one-step, chunked, absolute-joint, and compact diffusion policies are retained as negative baselines.

New statistical protocol. Five models are trained with seeds 42–46 on the same 4,000-frame dataset. Each is evaluated on 50 identical resets with random object pose, independent ± 2 cm translations of both cameras, half lighting, and a Gaussian background. We report mean and sample standard deviation over training seeds; paired resets support exact McNemar comparisons. A cumulative coverage ladder uses 1 demonstration, 500 clean frames, 2,000 frames with Gaussian/robot-pose coverage, 3,000 frames adding camera jitter, and 4,000 frames adding joint camera/lighting variation. A separate curve varies camera translation from 0 to 4 cm. No physical trial is included.

3. Results

3.1. Perception, geometry, and rendering

SAM/CoTracker passes the frozen perception gate: mean mask IoU is 0.777 for the moving source and 0.920 for the target, track survival is 91.4% and 95.1%, and there are no identity swaps. Correspondences are decisive for observable

Table 1. Single-image reconstruction after longest-axis metric alignment. Both learned meshes fail the 15% secondary-extent gate.

Geometry	Extents (mm)	Max err.	Watertight	VRAM
RGB-D reference	81.9/60.6/35.4	—	—	—
TripoSR-128	81.9/69.7/14.5	58.9%	yes	1.87 GB
SF3D	81.9/68.3/23.7	33.1%	no	6.17 GB

Table 2. Closed-loop controls on held-out resets. Small n rows are diagnostic; the selected policy receives the larger robustness study below.

Policy	Success	Parameters
Privileged state MLP	8/20	82k
Phase-conditioned image BC	0/5	335k
Dual-view action chunking	0/5	640k
Compact diffusion	0/5	944k
Absolute-joint chunking	0/5	643k
Temporal visual waypoint (ours)	19/20	2.61M

rotation: median held-out error falls by 98.0%, from 34.89 pixels (95% bootstrap CI 34.25–36.11) to 0.70 (0.57–0.91).

Table 1 shows that SF3D is proportionally closer but consumes the full device and is non-watertight; TripoSR is practical for visualization but flattens the unseen dimension. The metric path therefore uses RGB-D or primitives. The 7,007-Gaussian scene gains 0.49 dB mean PSNR on three held-out views after footprint tuning. Yet the tuned render lowers downstream success from 20/20 to 18/20, and retraining reaches 17/20. Image reconstruction quality therefore does not provide a policy-selection metric by itself. Metric registration yields 2.7 mm median table-plane residual and 99.2% depth-visible simulator foreground.

3.2. Closed-loop policy evidence

Table 2 reports architecture controls from 100 successful robosuite demonstrations. The compact direct visual policies all fail from held-out initializations despite low validation losses. The temporal waypoint interface instead succeeds in 19/20 trials. This result narrows the claim: learning stable spatial state and retaining an explicit controller works locally; it does not reproduce Video2Robo’s learned absolute-joint policy.

Across the new five-seed combined-shift experiment, mean success is **76.0±5.8%**, with seed outcomes spanning 68–82%. This uses 190/250 successful rollouts in total, but the seed distribution is the unit of replication. The mean falls below the frozen 80% combined-shift gate, revising the conclusion drawn from the originally selected checkpoint. That checkpoint attained 20/20 nominal, 19/20 camera, 20/20 half-light, 20/20 Gaussian-background, and 18/20 combined

successes; the repeated experiment shows that checkpoint selection hid material optimization variance.

The coverage ladder in Fig. 2 rises from 6% (one demonstration) and 2% (500 clean frames) to 20% after appearance and robot-pose coverage. Adding camera data produces 50%, while explicit combined-shift examples reach 82%; exact simulator geometry gives the 100% controller ceiling. This is evidence for coverage alignment rather than a generic “more data” effect because the stages change both sample count and variation type. The camera curve remains at 98–76% through the trained 0–2 cm range, then reaches 60% and 50% at 3 and 4 cm, locating an empirical viewpoint boundary rather than reporting one arbitrary perturbation.

3.3. Efficiency

The full local stack fits a GTX 1660 Ti. TripoSR peaks at 1.87 GB; SF3D peaks at 6.17 GB. Depth Anything V2 Small peaks at 415 MiB with 0.159 s median inference. Rendering the final 4,000-frame localization set is the main local data cost; training the 2.61M-parameter policy takes 55.4 s. One complete depth-ordered Gaussian robot demonstration contains 353 synchronized dual-view samples. These measurements complement Video2Robo’s L40 results and expose the accuracy lost when choosing small feed-forward geometry models.

3.4. Statistical contrasts and failure modes

Because every coverage variant sees the same 50 resets, we compare binary outcomes with two-sided exact McNemar tests. The 4,000-frame model beats the 3,000-frame camera model on 16 discordant episodes with no reverse discordance ($p = 3.05 \times 10^{-5}$). Its advantage over the 2,000-frame appearance/pose model is also decisive (32 gains and one loss, $p < 10^{-8}$). These paired tests do not prove that frame count is causal: the ladder intentionally adds examples from a new shift at each stage. They do show that the observed gains are not a consequence of different reset difficulty.

Failure traces connect success to perception. Under the combined shift, the full model’s 41 successful episodes have 4.47 mm median initial XY error; its nine failures have 11.90 mm. Eight of those failures terminate during the gripper-close phase, consistent with a missed grasp, while one reaches the final phase without satisfying placement. At 4 cm camera shift, successful and failed episodes separate further (8.12 versus 23.40 mm median error). Fourteen of 25 failures terminate at gripper close and nine fail during approach. The controller ceiling remains 50/50 under the combined condition, so the dominant measured bottleneck is visual localization rather than task physics.

The five-seed experiment exposes a second failure mode: model selection variance. All models use the same architecture, dataset, optimizer, and paired test resets, yet combined success spans 68–82%. Their median XY errors are much

Table 3. Scope comparison. “Ours” is designed to diagnose a small-machine reproduction and cannot support Video2Robo’s physical-transfer claim.

Dimension	Video2Robo	Ours
Tasks	6 rigid tasks	1 rigid Place task
Video input	Monocular RGB	RGB + RGB-D metric anchors
Geometry	learned scale/3DGS	measured primitives
Policy	Diffusion, joint targets	waypoint + phase control
Training GPU	NVIDIA L40	GTX 1660 Ti, 6 GB
Simulation	pose + multi-shift	five controlled conditions
Physical robot	Franka, two cameras	out of scope
Repeated seeds	not reported	5 training seeds

less variable (4.94–5.95 mm), indicating that a few tail predictions determine many grasp outcomes. Reporting only the best checkpoint therefore overstates reliability. The 95% t interval for mean seed success is 68.8–83.2%; it crosses the 80% gate but does not establish a pass. More seeds or a tail-aware training objective would be needed to narrow this uncertainty.

4. Relationship to Video2Robo

Table 3 states the intended comparison. Video2Robo evaluates six tasks and trains an absolute-joint Diffusion Policy from photorealistic robot videos. It reports 63.83% mean simulated success under multiple variation types, compared with 11.67–44.67% for its MimicGen/SkillGen controls, and performs physical Franka trials [4, 8]. Our higher task-specific percentages are not directly comparable: the task, policy interface, geometry source, and evaluation distribution differ. The useful reproduction question is whether the same qualitative factors—temporal tracking and frame-level scene diversity—remain important under a severe hardware budget. The ablations answer yes, while the seed study adds a robustness qualification absent from a single selected run.

The comparison also clarifies what Gaussian splatting contributes here. In Video2Robo, 3DGS represents editable task objects and the scene, supporting view-consistent robot data. In our retained policy dataset, simulator foreground is composited over an animated Gaussian background; metric registration is demonstrated separately but is not used to train the selected checkpoint. The experiment therefore tests Gaussian appearance coverage, not a complete 3DGS-based robotization pipeline. This narrower interpretation is necessary for an accurate report.

5. Discussion and Limitations

Three findings follow. **First, temporal correspondences are necessary.** Mask-only pose fitting is visually plausible yet inaccurate on an asymmetric object. **Second, visual reconstruction and metric reconstruction are different objectives.** Neither learned mesh supports the frozen colli-

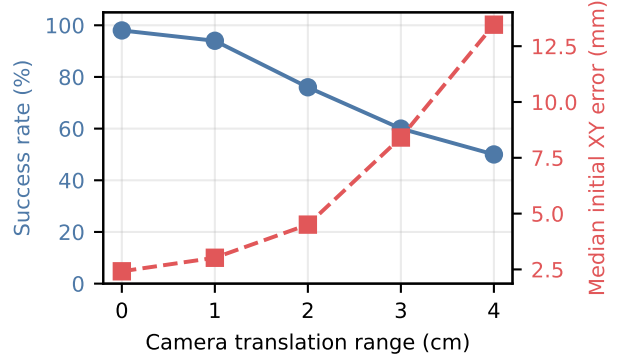
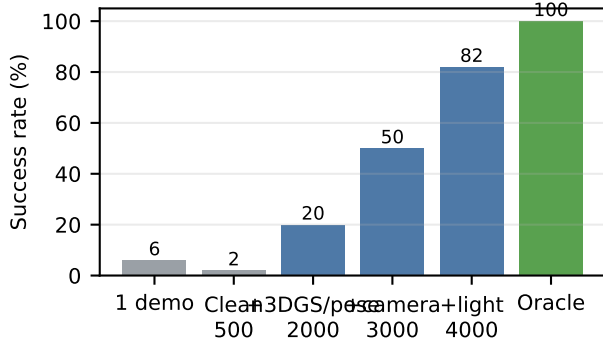


Figure 2. **Left:** paired combined-shift coverage ablation. Adding the shift observed at test time is more important than raw frame count. **Right:** success and initial localization error as camera translation exceeds the training range. Each point uses 50 identical resets.

sion gate, and higher held-out PSNR does not select the best controller input. **Third, data must cover the deployment shift, but coverage is not sufficient for seed-stable robustness.** Gaussian appearance alone does not repair camera geometry; explicit camera and low-light examples produce the largest combined-shift gain, while the five-seed mean still misses the 80% gate.

The study has clear boundaries. It contains one manipulation task, uses RGB-D scale and primitives rather than a monocular-only geometry estimate, composites Gaussian appearance around simulator foreground, and delegates phase execution to a scripted controller. The five training seeds share one generated dataset, so they measure optimization variability rather than source-video variability. No robot hardware is available; physical success and sim-to-real transfer remain unevaluated. Compared with Video2Robo’s six-task study and physical Franka trials, this report establishes a resource-constrained simulation result only.

6. Reproducibility and Future Work

The experiment runner freezes the 4,000-frame dataset, five training seeds, one paired reset seed, and 50 episodes per checkpoint. It resumes only from a report whose episode count matches the requested protocol, then regenerates JSON aggregates, LaTeX macros, and both plots. Raw observations, checkpoints, and episode traces stay outside version control because of their size; the committed manifest records their paths and the aggregate contains Wilson intervals, the seed distribution, and paired tests. This design makes the PDF a derived artifact of the same measurements rather than a manually transcribed summary.

Three extensions would materially change the evidence. First, repeat the entire pipeline on at least Place, Stack, and Sweep. These tasks add vertical placement and sustained contact, testing whether the waypoint interface is specific to a spherical pick-and-place task. Each task should use

an independent source video, five training seeds, and the same pose-only and multi-shift split. Second, compare the hybrid controller with an end-to-end action policy under an equal dataset and action horizon. Our compact diffusion negative result is diagnostic, but a tuned Diffusion Policy baseline is required before attributing the gap to policy class. Third, train from the metrically registered depth-ordered Gaussian sequence rather than appearance compositing. That would test the contribution of view-consistent 3DGS robotization and make the comparison to Video2Robo substantially closer.

The current camera curve also suggests a targeted algorithmic experiment. Errors grow smoothly from 2.42 mm at zero camera shift to 13.46 mm at 4 cm, while success falls from 98% to 50%. Training could sample a wider camera volume, but that may trade nominal precision for coverage. A factorial study should therefore cross camera range with calibration-aware inputs and report the whole severity curve, not tune on the 4 cm endpoint. Tail-aware objectives or ensembles may also reduce seed variance because median localization changes little across seeds while rare outliers determine grasp failure.

Finally, the monocular premise should be tested rather than assumed. A multi-object benchmark can compare learned depth/mesh scale, one manual dimension, RGB-D, and ground-truth simulator geometry using dimension error, ADD-S or BOP recall, collision validity, and downstream success. The present results predict that single-image mesh appearance will correlate weakly with metric fitness. Confirming that prediction across symmetric, thin, low-texture, and partly occluded objects would turn a task-specific engineering observation into a broader empirical result.

7. Conclusion

OneVideo2Policy reproduces a useful subset of the one-video synthetic-data pipeline on 6 GB VRAM. The strongest

evidence is not a claim of end-to-end Video2Robo equivalence, but a set of measured boundaries: correspondence tracking stabilizes motion, lightweight single-image meshes do not provide metric physics, and matched 3D appearance/camera/lighting coverage improves a compact learned waypoint policy but does not produce seed-stable combined-shift robustness. Broader tasks, source videos, and more stable training are the next hardware-free steps; physical validation requires a different study.

References

- [1] Mark Boss, Zixuan Huang, Aaryaman Vasishta, and Varun Jampani. SF3D: Stable fast 3D mesh reconstruction with UV-unwrapping and illumination disentanglement. *arXiv:2408.00653*, 2024.
- [2] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *RSS*, 2023.
- [3] Yinan Deng, Kejia Hu, Ye Chen, Jianyu Dou, Jiahui Wang, Jingyu Zhao, Haojia Ao, Yi Yang, and Yufeng Yue. Video2Robo: 3DGS-based synthetic data from one video enables scalable robot learning. In *CVPR*, 2026.
- [4] Caelan Garrett, Ajay Mandlekar, Bowen Wen, and Dieter Fox. SkillMimicGen: Automated demonstration generation for efficient skill learning and deployment. In *CoRL*, 2024.
- [5] Nikita Karaev, Iurii Makarov, Jianyuan Wang, Natalia Neverova, Andrea Vedaldi, and Christian Rupprecht. CoTracker3: Simpler and better point tracking by pseudo-labelling real videos. In *ICCV*, 2025.
- [6] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D gaussian splatting for real-time radiance field rendering. In *ACM SIGGRAPH*, 2023.
- [7] Yunze Liu, Haoran Jiang, Haoqiang Xu, Yizhou Wang, and Xinggang Wang. HOI4D: A 4D egocentric dataset for category-level human-object interaction. In *CVPR*, 2022.
- [8] Ajay Mandlekar, Soroush Nasiriany, Bowen Wen, Iretiayo Akinola, Yashraj Narang, Linxi Fan, Yuke Zhu, and Dieter Fox. MimicGen: A data generation system for scalable robot learning using human demonstrations. In *CoRL*, 2023.
- [9] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv:2408.00714*, 2024.
- [10] Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. TripoSR: Fast 3D object reconstruction from a single image. *arXiv:2403.02151*, 2024.
- [11] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. In *arXiv:2009.12293*, 2020.